

SEQUENTIAL SYSTEMS

Phase V: Data Cascades

(Shift Registers & Architectures)

Comprehensive Academic Master Manual

Optimized Visual Layout Edition:

Scaled Diagrams & Expanded Component Spacing

Strict Anti-Overlap Formatting

Methodology: Pure $\text{\LaTeX}$ Compilation

EE/PC/B/T/225 | Jadavpur University

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Chapter 1

The Architecture of Registers

1.1 Introduction to Data Storage

In earlier phases, we constructed flip-flops to trap and hold a single bit of information. We then engineered counters, which use internal logic to autonomously navigate through a sequence of mathematical states (e.g., 000, 001, 010).

However, a microprocessor requires hardware that can simply hold multi-bit data (like a user's password or an ALU calculation result) without automatically changing its state. An array of memory elements designed specifically for storing and moving data is called a **Register**.

An N -bit register is constructed from an array of N flip-flops (typically D-Flip-Flops) wired to a common master clock.

1.1.1 The Shift Register Mechanism

While a basic register simply stores data in parallel, a **Shift Register** is engineered with a specific physical topology: the output (Q) of one flip-flop is hard-wired directly into the data input (D) of the adjacent flip-flop.

Upon the arrival of an active clock edge, the data stored in the array cascades sideways, bit by bit, moving left or right across the silicon die.

Why Shift Registers are Essential

Shift registers are the absolute foundation of digital communication and arithmetic. They are used to convert Parallel data (8 wires simultaneously) into Serial data (1 wire flashing 8 times) for USB, Ethernet, and SATA transmissions. Furthermore, shifting a binary number left by one position mathematically multiplies it by 2, and shifting it right divides it by 2.

1.2 The Four Modes of Transfer

Shift Registers are classified by the temporal method used to load data into the array, and the method used to extract data from the array.

1.2.1 1. Serial-In, Serial-Out (SISO)

Data is injected into the first flip-flop exactly one bit at a time per clock pulse. To extract the data, it must be shifted out of the final flip-flop, one bit per clock pulse. It requires N clock pulses to load, and N clock pulses to read.

1.2.2 2. Serial-In, Parallel-Out (SIPO)

Data is injected serially, one bit per clock pulse (requiring N pulses). However, all internal Q outputs are physically tapped and wired to external pins. The entire stored word can be read simultaneously in a single instant.

1.2.3 3. Parallel-In, Serial-Out (PISO)

All N bits are jammed into the respective flip-flops simultaneously upon a single clock pulse using external combinational logic. The data is then extracted serially, one bit at a time, requiring N clock pulses to read.

1.2.4 4. Parallel-In, Parallel-Out (PIPO)

The fastest architecture. Data is loaded simultaneously across all bits, and read simultaneously across all bits. This is the standard architecture for high-speed

internal CPU registers (like the Accumulator).

Chapter 2

The Ultimate Architecture

2.1 The Universal Shift Register (IC 74194)

Instead of building four separate hardware architectures for SISO, SIPO, PISO, and PIPO, Intel and Texas Instruments engineers combined all functionalities into a single, deeply integrated silicon microchip: The **Universal Shift Register**.

The most famous industry-standard implementation is the 4-bit **IC 74194**. It permits bidirectional movement of data, as well as the capability to freeze data or load entirely new data simultaneously.

Its multifaceted operation is orchestrated by two **Mode Control Pins**: S_1 and S_0 .

2.1.1 The Multiplexer Routing Matrix

How does a single flip-flop know whether it is supposed to hold its data, receive data from its left neighbor, receive data from its right neighbor, or receive data

from an external parallel pin?

The 74194 solves this by placing a **4x1 Multiplexer (MUX)** immediately in front of the D-input of every single flip-flop. The mode control pins (S_1, S_0) act universally as the select lines for these four multiplexers.

By changing the binary value on the select lines, the hardware physically alters the routing network, dynamically connecting the D pin of the flip-flop to different wires.

2.2 The Mode Control Execution Table

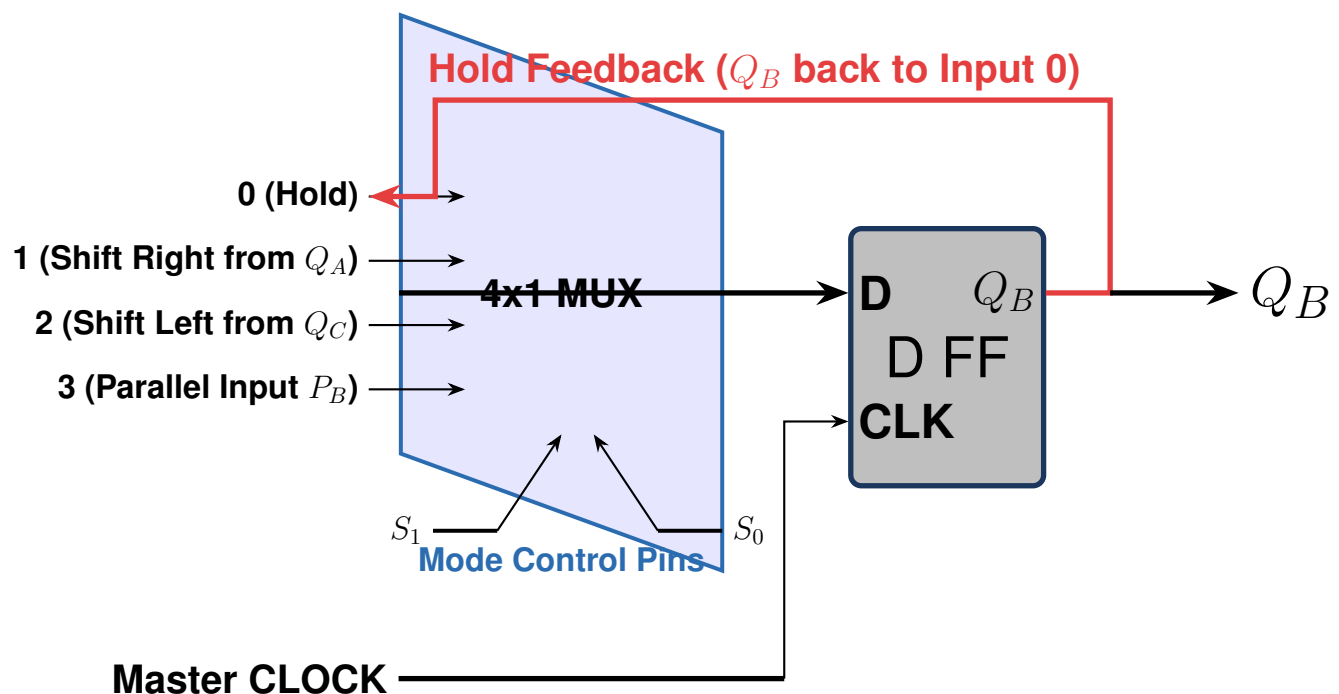
The logic table below dictates exactly how the multiplexers route the signals based on the active mode selected by the engineer.

The Power of Synchronicity
Because the 74194 routes all multiplexed signals directly into the D-inputs of edge-triggered flip-flops sharing a single master clock, every operation (Hold, Shift, or Load) occurs strictly on the positive edge of the clock pulse. There is no asynchronous ripple.

Control		Universal Register Operation	
S_1	S_0	Active Mode	MUX Routing Mechanics
0	0	HOLD	MUX selects input 0. The Q output of each flip-flop is routed straight back into its own D input. The clock pulses, but the state does not change.
0	1	SHIFT RIGHT	MUX selects input 1. Each flip-flop receives data from its left-hand neighbor ($Q_A \rightarrow Q_B \rightarrow Q_C \rightarrow Q_D$).
1	0	SHIFT LEFT	MUX selects input 2. Each flip-flop receives data from its right-hand neighbor ($Q_D \rightarrow Q_C \rightarrow Q_B \rightarrow Q_A$).
1	1	PARALLEL LOAD	MUX selects input 3. Each flip-flop is disconnected from its neighbors and receives data directly from the external parallel input pins.

2.3 Architectural Schematic of the 74194

The diagram below illustrates the routing architecture for a single stage (Bit B) of the 4-bit Universal Shift Register. Multiply this structure four times to build the complete IC.



*** END OF PHASE V MASTER
MANUAL ***